

Final

$$\textcircled{1} \quad A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 3 & 9 \end{bmatrix}$$

$$i) \quad A = LU$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

L U

ii) $\text{rref}(A)$

$$R = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

iii) Basis for $N(A)$

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{iv) } Ax = b$$

$x \rightarrow$ particular solution + nullspace

$$x = p + \lambda \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Since b is given as sum of columns of A .

$$b = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\underline{\textcircled{2}} \quad y = c + d 2^t$$

$$(t, y) = (0, 6) \quad (1, 4) \quad (2, 0)$$

$$\begin{aligned} \text{a)} \quad 6 &= c + d \\ 4 &= c + 2d \\ 0 &= c + 4d \end{aligned}$$

$$\text{b)} \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix}$$

$$\begin{aligned} A^T A &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 7 \\ 7 & 21 \end{bmatrix} \end{aligned}$$

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 3 & 7 \\ 7 & 21 \end{bmatrix} \hat{x} = \begin{bmatrix} 10 \\ 14 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 3 & 7 & 1 & 0 \\ 7 & 21 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} 3 & 7 & 1 & 0 \\ 0 & 14 & -7 & 3 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} 3 & 0 & 1 + 7/2 & -3/2 \\ 0 & 14 & -7 & 3 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 3/2 & -1/2 \\ 0 & 1 & -1/2 & 3/4 \end{array} \right]$$

$$\begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 3/4 \end{bmatrix} \begin{bmatrix} 10 \\ 14 \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \end{bmatrix}$$

c) $A^T b = 0$ as $\hat{c} \hat{d} = [0 \ 0]$

$$\begin{bmatrix} 1 & 1 & 1 \\ & 1 & 3 \end{bmatrix} b = 0$$

$$\boxed{N(A^T) = \text{values of } y}$$

$$\textcircled{3} \quad A v_i = b_i$$

$$a) \quad AV = B$$

If all the v_i 's are independent
then A can be determined uniquely.

$$\boxed{A = BV^{-1}}$$

$$b) \quad C(A) = k_1 b_1 + k_2 b_2 \dots k_n b_n$$

c) If all the b_i are linearly
independent then we can find
 A^{-1}

$$A^{-1} = VB^{-1}$$

④ $x_k \rightarrow$ linear
 $1-x_k \rightarrow$ catalase

$$a) \quad x_{k+1} = \frac{4}{5} x_k + \frac{1}{10} y_k$$

$$y_{k+1} = \frac{9}{10} y_k + \frac{1}{5} x_k$$

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} 4/5 & 1/10 \\ 1/5 & 9/10 \end{bmatrix} \begin{bmatrix} x_k \\ y_k \end{bmatrix}$$

$$A = \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix}$$

$$\lambda = 1$$

$$\lambda = 0.7$$

$$X_k = c_1 (\lambda_1)^k x_1 + c_2 (\lambda_2)^k x_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$c_1 = \frac{1}{3}$$

$$c_2 = \frac{2}{3}$$

$$X_{\infty} = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$$

$$b) \quad x(0) = 1 \quad y(0) = 0$$

$$\frac{dx}{dt} = 3x - 4y$$

$$\frac{dy}{dt} = 2x - 3y$$

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix} = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\frac{dx}{dt} = Ax$$

$$x = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2$$

$$\begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} \quad x_0 = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = \pm 1$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(-1)

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(1)

$$\boxed{\begin{matrix} c_1 = -1 \\ c_2 = 1 \end{matrix}}$$

$$X = e^t \begin{bmatrix} 2 \\ 1 \end{bmatrix} - e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 2e^t - e^{-t} \\ e^t - e^{-t} \end{bmatrix}$$

c) What X_0 will make $\begin{bmatrix} x_t \\ y_t \end{bmatrix}$ lie on a strg line.

→ When y can be linearly expressed
in x

$$\text{So, } \begin{aligned} x_t &= 2c_1 e^t + c_2 e^{-t} \\ y_t &= c_1 e^t + c_2 e^{-t} \end{aligned}$$

$$c_1 = 0 \quad \text{OR} \quad c_2 = 0$$

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \end{bmatrix} / k \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(5) 120° $x = y = z$ $i = (1, 0, 0)$
 $\downarrow$
 $j = (0, 1, 0)$
 $\downarrow$
 $k = (0, 0, 1)$

Projection on line $c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$\# P = \frac{aa^T}{a^T a} = \frac{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}}{3} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P_i = i - P_i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

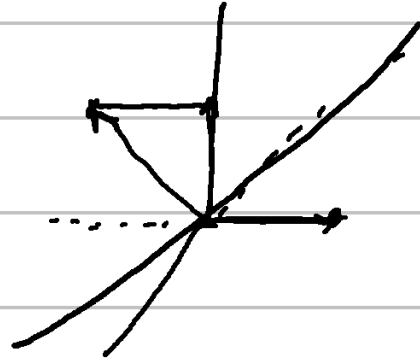
$$P_i = \frac{1}{3} \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} \quad P_j = \frac{1}{3} \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} \quad P_k = \frac{1}{3} \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$P_i \cdot P_j = \frac{1}{9} (-2 - 2 + 1) = -\frac{1}{3}$$

$$= \|P_i\| \|P_j\| \cos \theta = \frac{2}{3} \cos \theta = -\frac{1}{3} \quad \boxed{\cos \theta = -\frac{1}{2}}$$

$$\text{Hence } \theta = 120^\circ$$

$$j - i = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$



perpendicular to axis.

$$b) \quad A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$A^3 = I$ because this is shift matrix everything shifted 3 times would return to same value.

$$(A - \lambda I) = \begin{bmatrix} -\lambda & 0 & 1 \\ 1 & -\lambda & 0 \\ 0 & 1 & -\lambda \end{bmatrix}$$

$$-\lambda(\lambda^2) + 1(1) = 0$$

$$\lambda^3 = 1, e^{2\pi i/3}, e^{4\pi i/3}$$

c) $P \rightarrow$ projection onto $\underline{x + 2y + z = 0}$

$P \rightarrow 0, 1, 0$

plane perpendicular
to vector $(1, 2, 1)$

$$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

6

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$

$$\begin{aligned} a) \quad A^T A &= \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix} \quad \boxed{\lambda = 0, 70} \end{aligned}$$

$$\begin{bmatrix} -56 & 28 \\ 28 & -14 \end{bmatrix}$$

$$\begin{aligned} -4x + 2 &= 0 \\ x &= \frac{1}{2} \end{aligned}$$

λ	0	70
x	$\begin{bmatrix} -2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$AA^T = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 10 & 15 \\ 10 & 20 & 30 \\ 15 & 30 & 45 \end{bmatrix} \quad \lambda = 0, 0, 70$$

$$\begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} \quad \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -65 & 10 & 15 \\ 10 & -50 & 30 \\ 15 & 30 & -25 \end{bmatrix}$$

$$\begin{bmatrix} -9 \\ 20 \\ -3 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} \quad \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$-9a + 20b - 3c = 0 \rightarrow -30c + 20b = 0$$

$$a - 3c = 0$$

$$3c = 2b$$

$$a = 3c$$

$$\begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$$

$$\boxed{c=2 \text{ if}}$$

$$\begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} \quad \begin{bmatrix} -9 \\ 20 \\ -3 \end{bmatrix} \quad \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$$

Differs from
the answer
but i think
it's a choice

b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad A_i = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Component of b that is $\perp$ to A_i
is non-existent.

c) $A_{m \times n}$ $m > n$ $\text{rank} \leq n < m$

$AA^T \rightarrow m \times m \rightarrow$ singular so
positive semidefinite

$A^T A \rightarrow n \times n \rightarrow$ not always
positive definite
could be zero

⑦ a) $\det A = 0$ as singular matrix.

$$b) C_{11} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = -1$$

$$\det B = -1 C_{11} = \textcircled{1}$$

after subtracting row 2 from row 1

This is value enclosed by

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$c) \det C = (1-x) C_{11} = \underline{1-x}$$

⑧ a) $B = M^{-1} A M$

$$A^k \rightarrow 0 \quad \text{then} \quad B^k \rightarrow 0$$

$k \rightarrow \infty$

$$B^k = M^{-1} A^k M$$

Hence when $k \rightarrow \infty$ & $A \rightarrow 0$

$$\underline{\underline{B^k = M^{-1}(0)M \Rightarrow 0}}$$

b) S fixed invertible 3×3 that diagonalises A . $S^{-1}AS$ is diagonal

$A \rightarrow$ subspace.

$$\Rightarrow S^{-1}(cA)S \rightarrow c(S^{-1}AS) \checkmark$$

$$S^{-1}(A+B)S \rightarrow \text{diagonal}$$

$S^{-1}AS + S^{-1}BS$

diagonal diagonal

c) 3×3 diagonal matrix basis.

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Basis for b

$$S^T A S = \Lambda_1$$

$$S^T B S = \Lambda_2$$

$$A = S \Lambda_1 S^T$$

$$B = S \Lambda_2 S^T$$

same eigenvectors

But different eigenvalues.

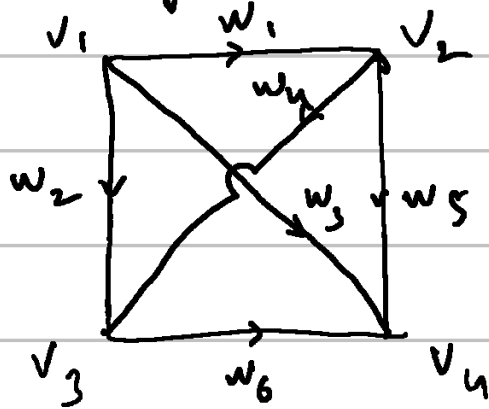
$$S^T \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} S, \quad S^T \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} S,$$

$$S^T \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} S$$

②

4 nodes 6 edges

voltage $\rightarrow (v_1, v_2, v_3, v_4)$



a)

$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}_{6 \times 4}$$

$$b) A^T A =$$

$$\begin{bmatrix} -1 & 0 & -1 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}_{6 \times 4}$$

$$\Rightarrow \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix} = A^T A$$

$$\text{col } 1 + \text{col } 2 + \text{col } 3 + \text{col } 4 = 0$$

$$\text{rank}(A^T A) = 3$$

$$N(A^T A) = c \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c) A^T A v = 0$$

$$A^T A \begin{bmatrix} 1 \\ v_2 \\ v_3 \\ 0 \end{bmatrix} = 0$$

$$\begin{bmatrix} 3 - v_2 - v_3 \\ -1 + 3v_2 - v_3 \\ -1 - v_2 + 3v_3 \\ -1 - v_2 - v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v_1 \neq v_{ex}$$

$$v_2 + v_3 = 3$$

$$v_3 - 3v_2 = 1$$

No solution !!